low electron speeds, a resistance of ⅔ the same as he had computed before; at high speeds, a resistance of ⅓.

By combining his ⅓ resistance for *relativistically degenerate matter* (that is, matter so dense that the degenerate electrons move at nearly the speed of light) with the formulas given in Eddington's book, Chandrasekhar then deduced the properties of high-density, high-mass white dwarfs. The answer was astonishing: The high-density matter would have difficulty supporting itself against gravity—sufficient difficulty that *only if the star's mass were less than that of 1.4 Suns could the squeeze of gravity be counterbalanced* This meant that no white dwarf could ever have a mass exceeding 1.4 solar masses!

With his limited knowledge of astrophysics, Chandrasekhar was deeply puzzled about the meaning of this strange result. Time and again Chandrasekhar checked his calculations, but he could find no error. So, in the last few days of his voyage, he wrote two technical manuscripts for publication. In one he described his conclusions about the structure of low-mass, low-density white dwarfs such as Sirius B. In the other he explained, very briefly, his conclusion that no white dwarf can ever be heavier than 1.4 Suns.

When Chandrasekhar arrived in Cambridge, Fowler was out of the country. In September, when Fowler returned, Chandrasekhar eagerly went to his office and gave him the two manuscripts. Fowler approved the first one and sent it to *Philosophical Magazine* for publication, but the second one, the white-dwarf maximum mass, puzzled him. He could not understand Chandrasekhar's proof that no white dwarf can be heavier than 1.4 Suns; but then, he was a physicist rather than an astronomer, so he asked his colleague, the famous astronomer E. A. Milne, to look at it. When Milne couldn't understand the proof either, Fowler declined to send it for publication.

Chandrasekhar was annoyed. Three months had passed since his arrival in England, and Fowler had been sitting on his paper for two months. This was too long to wait for approval to publish. So, piqued, Chandrasekhar abandoned his attempts to publish in Britain and mailed the manuscript to the *Astrophysical Journal* in America.